

Jared Leonard

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PROFILE

Computer Science (BSc Hons, 2:1) graduate with experience designing and implementing backend services, APIs, and data-driven systems. Strong foundation in software engineering principles including API design, data modeling, testing, and debugging, with hands-on experience deploying and operating services in cloud environments. Interested in software engineering roles with exposure to scalable systems and real-world domains.

TECHNICAL SKILLS

Programming

- Python (FastAPI), Java (OOP, collections), C (systems programming), Dart
- SQL (schema design, normalization), Bash

Software Engineering

- REST API design and request lifecycle
- Data modeling and persistence
- Input validation, error handling, and testing
- Debugging, refactoring, and documentation
- Git workflows and code reviews

Systems and Cloud

- AWS (Lambda, DynamoDB, S3, IAM)
- Docker, Terraform (Infrastructure as Code)
- CI/CD with GitHub Actions
- Linux fundamentals, concurrency, TCP/IP sockets
- Observability basics: logging, health checks

Datastores

- PostgreSQL, MySQL, DynamoDB, MongoDB, Supabase

EDUCATION

TU Dublin

BSc in Computer Science (Honours)

Classification: 2:1

Dublin, Ireland

2021 - 2025

Kyungpook National University

BSc in Engineering (Erasmus)

Classification: G.P.A - 3.10/4.3

Daegu, South Korea

2024-2024

PROJECTS

BrewHand — Mobile App

- Designed and implemented an end-to-end mobile application with realtime backend integration.
- Built asynchronous client-backend interactions to handle concurrent user actions and realtime updates.
- Modeled application data using a relational schema and managed persistence via a hosted backend.
- Structured frontend and backend logic into modular components to support maintainability and feature expansion.
- Integrated authentication, realtime subscriptions, and backend APIs into a cohesive production-style system.
- **Tech:** Flutter, Dart, Supabase, PostgreSQL
- **Repo:** github.com/jaredleoa/Brewhand

Transaction Risk Scoring Service — FinTech Platform Engineering

- Designed and implemented a backend REST API that evaluates transaction risk based on amount, country, and merchant rules.
- Built deterministic scoring logic with structured validation and error handling using FastAPI.
- Modeled transaction inputs and responses to support stateless execution and horizontal scalability.
- Wrote unit tests with pytest to validate scoring behavior and API correctness.
- Containerised the service with Docker and automated builds and tests using GitHub Actions.
- **Tech:** Python, FastAPI, Docker, Pytest, GitHub Actions
- **Repo:** github.com/jaredleoa/transaction-risk-service

Payments Event Pipeline — Cloud Systems

- Implemented a payment event ingestion pipeline writing structured events into DynamoDB.
- Developed AWS Lambda logic to validate and persist incoming payment records.
- Designed reproducible cloud infrastructure via Terraform with managed state, variables, outputs.
- Verified end-to-end behavior through controlled test events and AWS CLI inspection.
- **Tech:** Python, Terraform, AWS (Lambda, DynamoDB, S3, IAM)
- **Repo:** github.com/jaredleoa/payments-pipeline

Secure File-Transfer Server — Systems Software

- Implemented a multi-threaded POSIX socket server supporting authenticated file transfer.
- Used mutexes to protect shared resources and prevent race conditions under concurrent access.
- Stress-tested with multiple clients to validate correctness and stability.
- **Tech:** C, POSIX sockets, pthreads

Distributed Marketplace — Distributed Systems

- Built a distributed buyer-server architecture handling concurrent requests and shared state.
- Designed a custom application-level protocol with graceful disconnect and failure handling.
- **Tech:** Python, sockets, threading.

WORK EXPERIENCE

Sureskills

Dublin, Ireland

Media Specialist

2021 - Present

- Maintained and improved technical IT content across 40+ courses.
- Executed version-controlled update workflows, release documentation, and technical validations.

Starbucks

Dublin, Ireland

Supervisor

2023 - Present

- Managed operational workflows in a €1.5M annual-revenue store.
- Led a team of 15, controlled deployment, and resolved incidents under time pressure.
- Built and implemented a fairness-based staff task-allocation system.

CERTIFICATIONS

- AWS Certified Cloud Practitioner — In progress (2025)

ADDITIONAL

- **Korean:** TOPIK Level 2 (165/200)200. Upper Intermediate.
- **Interests:** cloud infrastructure, distributed systems, payments engineering